BC7.Applying Critical Thinking and Problem Solving Techniques in the Workplace

Visual Graphic Design (Southern Luzon Technological College Foundation)

COMPETENCY-BASED LEARNING MATERIAL

Sector:	Information and Communications Technology
Qualification Title:	Visual Graphic Design NCIII
Unit of Competency:	Apply Critical Thinking and Problem Solving Techniques in the Workplace
Module Title:	Applying Critical Thinking and Problem Solving Techniques in the Workplace

MACO INSTITUTE OF TECHNOLOGY, INC.
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HOW TO USE THIS COMPETENCY-BASED LEARNING MATERIAL

This unit of competency, “Apply Critical Thinking and Problem Solving Techniques in the Workplace”, is one of the competencies of Visual Graphic Design NCIII, a course which comprises the knowledge, skills, and attitudes required for a TVET trainee to possess.

This module, Applying Critical Thinking and Problem Solving Techniques in the Workplace, contains training materials and activities related to applying parts of speech; applying sentence structure; communication skills; math basic; and introduction to technical writing.

In this module, you are required to go through a series of learning activities in order to complete each learning outcome. In each leaning outcome are *Information Sheets, Self-Checks, Operation Sheets, Task Sheets, and Job Sheets.* Follow and perform the activities on your own. If you have questions, do not hesitate to ask for assistance from your facilitator.

Remember to:

- Read the Information Sheets and complete the Self-Checks
- Perform the Task Sheets, Operation Sheets and Job Sheets until you are confident that your outputs conform to the Performance Criteria Checklists that follow the said work sheets.
- Submit outputs of the Task Sheets, Operation Sheets and Job Sheets to your facilitator for evaluation and recording in the Achievement Chart. Outputs shall serve as your portfolio during the Institutional Competency Evaluation. When you feel confident that you have had sufficient practice, ask your trainer to evaluate you. The results of your assessment will be recorded in your **Achievement Chart** and **Progress Chart.**

You must pass the Institutional Competency Evaluation for this competency before moving to another competency. A Certificate of Achievement will be awarded to you after passing the evaluation.
You need to complete this module in order to go through the next module.

VISUAL GRAPHIC DESIGN NCIII
COMPETENCY-BASED LEARNING MATERIALS

LIST OF COMPETENCIES

No.	Unit of Competency	Module Title	Code
1.	Lead Workplace Communication and Observe Gender Sensitivity	Leading Workplace Communication and Observe Gender Sensitivity	500311109
2.	Lead Small team	Leading small team	500311110
3.	Develop and practice negotiation skills	Developing and practicing negotiation skills	500311111
4.	Solve workplace problem related to work values and gender sensitivity	Solving workplace problem related to work values and gender sensitivity	500311112
5.	Use mathematical concepts and techniques	Using mathematical concepts and techniques	500311113
6.	Use relevant technologies	Using relevant technologies	500311114
7.	***Apply critical thinking and problem solving techniques in the workplace***	***Applying critical thinking and problem solving techniques in the workplace***	***500311142***
8.	Use information creatively and critically	Using information creatively and critically	500311144
9.	Work in a diverse environment	Working in a diverse environment	500311145

UNIT OF COMPETENCY **:** APPLY CRITICAL THINKING AND PROBLEM SOLVING TECHNIQUES IN THE WORKPLACE

MODULE TITLE **:** APPLYING CRITICAL THINKING AND PROBLEM SOLVING TECHNIQUES IN THE WORKPALCE_

MODULE DESCRIPTION : This unit covers the knowledge, skills and attitudes required to solve problems in the workplace including the application of problem solving techniques and to determine and resolve the root cause of problems.

NOMINAL DURATION : 8 hours

QUALIFICATION LEVEL NC III

SUMMARY OF LEARNING OUTCOMES:

Upon completion of the module, the learner/students must be able to:

LO1. Identify the problem.

LO2. Determine fundamental causes of the problem

LO3. Determine corrective action

LO4. Provide recommendation/s to manager

LO1. IDENTIFY THE PROBLEM

ASSESSMENT CRITERIA:

1. Variances are identified from normal operating parameters; and product quality
2. Extent, cause and nature of the problem are defined through observation, investigation and analytical techniques.
3. Problems are clearly stated and specified.

CONTENTS:

- Process, normal operating parameters and product quality to recognize nonstandard situations.
- Enterprise goals, targets and measures
- Analytical techniques
- Types of problems.

CONDITION:

The students/learners must be provided with the following:

- Hand-outs

METHODOLOGIES:

- Lecture
- Group discussion

ASSESSMENT METHODS:

- Written Examination
- Oral Evaluation

INFORMATION SHEET 7.1-1
NON-STANDARD WORK AND WORKERS

How do nonstandard work arrangements affect employees and organizations?

Much of the research on nonstandard workers has focused on the attitudes and behaviours of nonstandard workers, especially as compared to standard workers. This stream of work has, for the most part, been based on the assumption that nonstandard workers on account of their limited temporal, physical or administrative attachment to organizations demonstrate weaker attachment to the organization.

This weaker attachment is argued to be manifested in the form of fewer citizenship behaviours (Ang & Slaughter, 2001), lower performance (Belous, 1989), lower identification with the organization (Wiesenfeld, Raghuram, & Garud, 1999), lower job satisfaction (Hall, 2006; Miller & Terborg, 1979) or lower commitment to the organizations (Van Dyne & Ang, 1998). However, the literature has not provided uniform support for these arguments. While some studies have found that nonstandard workers are less attached to the organization (Van Dyne & Ang, 1998; Forde & Slater, 2006; Hall & Gordon, 1973), others have found no differences between standard and nonstandard workers (Haden, Caruth, & Oyler, 2011; Pearce, 1993), and still others have found that nonstandard workers are more attached to organizations than their standard colleagues (De Cruyper & De Witte, 2007; Eberhardt & Shani, 1984; Galup, Saunders, Nelson & Cerveny, 1997; Katz, 1993; McDonald & Makin, 2000; Parker et al., 2002).

Most of these studies have been done using samples of temporary workers who have limited temporal connections with the organization. However, the same logic that limited exposure to the organization would decrease the attachment of remote workers or contract workers was proposed and tested by researchers. As was the case for temporary workers, the findings were again mixed with these types of nonstandard workers also not all uniformly responding more negatively to the organization than comparable standard workers (e.g. Pearce, 1993).

Researchers have tried to understand these mixed results by examining if the differences in attitudes and behaviours are caused perhaps not merely by the work arrangement but also by factors such as the extent to which the individual nonstandard worker has chosen this work arrangement (Ellingson et al., 1998), or the type of tasks they undertake (Ang & Slaughter, 2001) or indeed the nature of their employment arrangements (Chambel & Castanheira, 2012). For instance, one set of arguments was that individuals who chose to work on reduced hours or flexible hours would be more positively inclined towards organizations that facilitate their working in this way (Holtom, Lee & Tidd, 2002).

These arguments were also consistent with the focus on the boundaryless career (Marler, Barringer & Milkovich, 2002) which made the point that individuals' careers involve movement in and out of organizations, and that nonstandard work might be a temporary arrangement that is desired by the worker. There is some empirical support for these arguments in that individuals who work in arrangements of their choice are more positively inclined to the organization, than individuals who do not (Holtom et al.,2002; Tan & Tan, 2002).

However, there are no quantitative studies that have followed individuals over their careers to study movement in and out of formal standard organizational work arrangements. Also, there is some empirical evidence that the type of job that a person does affects their attachment to the organization. Individuals who have more autonomous jobs are more attached to the organization, even when they are in a temporary position (Ang & Slaughter, 2001; 10 Conditions of Work and Employment Series No. 61 Hundley, 2001). Chambel and Castanheira (2012) argued that there is a social exchange process that underlies workers' attachment to organizations. They found in a sample of Portuguese blue-collar workers from a temporary help agency that when organizations provided training to these workers they reciprocated by reporting high affective commitment to the organization. More recently, Van Jaarsveld & Liu (2015) found in a study of call-centre workers in China that when workers experienced low involvement practices in the workplace the turnover in the workplace was high.

A second broad stream of work that has examined the effect of nonstandard work arrangements has examined the consequence to workers of being in a blended workforce or work group. This stream of research arose from the observation that contrary to the core-periphery hypothesis (Piore & Deoringer, 1971), standard and nonstandard workers are not segregated from each other in the workplace (Davis-Blake et al., 2003). Rather they work alongside each other, often in similar jobs, and this contact is likely to make salient to them the different terms of employment (Chattopadhyay & George, 2001). For instance, by working alongside each other they would be aware of different wages, different levels of job security and different benefits. Researchers like George (2003) and Davis-Blake et al., (2003) examined whether the proportions of nonstandard workers in a department or workgroup affected the attitudes of standard workers towards the organization and towards their co-workers.

These researchers found that the greater the proportion of nonstandard workers, the more negative the standard workers' attitudes. They argued that the reason for this was that the presence of nonstandard workers signalled the management's decreasing intention of investing in its workforce and consequently standard workers started worrying about the security and value of their own jobs. More recently, George, Chattopadhyay & Zhang (2012) found that standard workers who believe that nonstandard workers cannot move up the organizational hierarchy (and thus threaten their jobs) perceive their nonstandard colleagues to be helping hands rather than competition. Under these conditions standard workers respond positively to working with nonstandard workers.

Chattopadhyay & George (2001) also found that the lower the proportion of nonstandard workers in the workgroup was, the more positive were the attitudes of the temporary workers in the workgroup. This they suggested was because temporary workers view the opportunity to work with standard workers positively, while if they had more nonstandard colleagues they would view their work team as peripheral to the organization. In summary, this body of research suggests there is a social comparison process which influences how workers perceive their work arrangements. If they feel valued and secure in their jobs they are more likely to be positively inclined towards their coworkers and the organization.

This research also highlights the social exchange process where individuals who feel short-changed by the organization reciprocate by decreasing their commitment to the organization. Individuals' perceptions are key to predicting their responses to nonstandard work arrangements. Consequently, how management communicates its intent to all workers would be critical in managing expectations related to nonstandard work arrangements and their effect on workers.

How has this affected the management of human resources in organizations?

In this section we will consider two key issues that are relevant to organizations. The first is how to manage workers when you have both standard and nonstandard workers in the same workplace. There is a small but growing body of research on this topic that we will discuss with a view to understanding its implications for competencies that the organization must have to manage nonstandard workers and blended workgroups.

The second issue is how organizational capabilities are affected by nonstandard work. This is a more difficult issue to explore empirically and has not been directly studied by researchers. Nevertheless we will use other studies from human resource management more generally to extrapolate on how the use of nonstandard workers might affect the managerial competencies of organizations.

Managing Nonstandard Workers

There are at least four ways in which organizations can think of building a connection with the nonstandard workers that they employ (Ashford et al., 2008). These include the design of jobs that are suitable for nonstandard workers, managing the terms of exchange between the firm and these workers, managing relationships of nonstandard and standard workers in the workplace and aligning the interests of the organization with those of the nonstandard workers. We discuss these below.

Managing Job Design: Organizations can manage their nonstandard workforce by carefully designing jobs that are amenable for workers who do not work full-time or on location, or are employees of a third firm. Typically, these are jobs with less complexity or that demand fewer or lower levels of skills (Davis-Blake & Uzzi, 1993), or jobs that are linked to seasonal demands of products or services (Houseman, 1997). The key competency required of the organization is ability to match jobs with the individuals in the job. For example, Cascio (2000) argued that virtual work is not appropriate for individuals new to a task, but rather a better fit when they are comfortable with the tasks and the organization. Similarly, assigning complex tasks that require working interdependently with others will not work for temporary workers who are with the organization for a short period of time (Sias, Kramer & Jenkins, 1997).

Managing the Terms of Exchange: The employment relationship is one that is defined by exchange between individuals and organizations. This would suggest that the management of nonstandard workers entails having a good understanding of what these nonstandard workers consider as fair or appropriate terms of exchange. Liden, Wayne and Kraimer (2003) found that contingent workers are more committed to organizations that provide them support and treat them fairly. Barley & Kunda (2004) found that contract workers prefer organizations that allow them to stay out of the politics of the workplace. Both Tan and Tan (2002) and Van Dyne and Ang (1998) found in studies of temporary workers in Singapore that temporary workers in a tight labour market are willing to work harder that their standard colleagues in the anticipation that their work will be rewarded with permanent contracts. These studies indicate that there is heterogeneity in what workers want and that the social exchange is affected by the labour market at a point in time.

Managing Interpersonal Relationships: Nonstandard workers are often in a liminal position trying to manage a position that is both within and outside the organization. Thus one way in which to manage them is through managing the nature of their relationships with people in the workplace. There are multiple reasons why nonstandard workers are excluded from social networks at work. These include their physical distance 12 Conditions of Work and Employment Series No. 61 from others as is the case for virtual workers (Wiesenfeld et al., 1999), or because they are temporary (Rogers, 2000; Wheeler & Buckley, 2000), or because of organizational practices such as making temporary workers wear distinctive badges and uniforms that highlight the difference between them and standard workers (Smith, 1998). However, a number of studies have shown that good interpersonal relationships with co-workers (Chattopadhyay & George, 2001), or supervisors (Benson, 1998) can improve contract workers and temporary workers' identification with their client organizations. The managerial competency required for the effective management of nonstandard workers is one of developing processes that facilitate good horizontal and vertical interpersonal relationships.

Managing Identity: A small set of studies show that managers can connect temporary or contract workers to the organization by helping these workers see how their identities are aligned with that of the organization (Chattopadhyay & George, 2001; George & Chattopadhyay, 2005). These studies show that when workers can see how their personal interests, and the ways in which they view themselves, can be facilitated by the organization they are more committed to the organization and its' interests. Barley and Kunda's (2004) research on knowledge workers suggests that these workers are happy to be outside of organizations, or "hired guns" since they see this liminal position as consistent with their own professional identities. Similarly, George & Chattopadhyay's (2005) study of leased workers in the IT industry show that these workers identified with more prestigious firms, since the positive image of the organization helped facilitate their own positive views of themselves. The managerial competency that this research suggests is of understanding the core of nonstandard workers' identities (and their related motivations) and engaging with them in a way that can help these workers realize, maintain or enhance these identities.

Effect of Nonstandard Work Arrangements on HRM competencies.

Nonstandard work arrangements change the ways in which organizations can, and do, manage their human resources. Right from the decision on whether to have the work done in-house with standard workers, or to hire workers on short term contracts, or to outsource the work to an external agency that manages contract workers, nonstandard work affects basic human resource management practices such as employee selection, training and skill development, planning and retention. We discuss these below.

Recruitment and Selection: The use of nonstandard workers can have multiple effects on the recruitment and selection functions of organizations. Since nonstandard work is often associated with tasks that are not core to the organization (Shi, 2007), or that do not involve valuable and proprietary knowledge or technology (Mayer & Nickerson, 2005), the recruitment and selection of workers to these positions does not require the care and precision that would be required for hiring workers who would move into longer term contracts in jobs that are central to the competitiveness of the organization (Lepak and Snell, 2002). Further, short term contracts enable organizations to "try out" workers before they move them into more permanent positions, involving longer term contracts (Bauer, & Truxillo, 2000). The use of nonstandard work arrangements is then seen as a way to reduce the uncertainty in the selection process. If firms can try before they hire, they can be sure that the individuals they finally offer standard contracts to as indeed the ones who add the most value to them. Nevertheless, this dependence on external sources to find employees could reduce the organization's capability of recruiting from the external market, and the practice of prolonging the selection process, as it were, could reduce the organization's selection capability. The effect of using temporary workers on the recruitment and selection capability of organizations has however not been studied so this is clearly an area that needs further investigation. Conditions of Work and Employment Series No. 61 13

Training and Skill Development: The use of nonstandard work arrangements has to a large extent shifted the responsibility of training and development from organizations to individual workers (Barley and Kunda, 2004). As a result, the greater the proportion of nonstandard workers in an organization there is a commensurate decrease in the organizational investment in the training of these employees (Davis-Blake & Uzzi, 1993). One of the implications of these work arrangements is that the HR function shifts its competencies from training and development of employees within the organization to identifying the sets of skills they need to buy from the markets and procuring these skills in an efficient and timely manner. This shift requires the organization to have good HR systems that facilitate the timely recognition of the needs for particular types of skills or competencies in the organization. At the same time this dependence on buying all the skills that the firm needs could affect organizations in two ways. First, it could result in a gradual erosion of firm-specific skills in the organization (Lepak &

Snell, 2002). Organizations that describe their human resources as one of their key assets then have assets that are not very distinct from that of their competitors, thus diminishing the role of people as a source of competitive advantage. A second implication of the use of temporary or contract workers rather than training employees is that the firms' ability to respond to changing markets might be restricted. Since the focus is less on training-for-skills and more on hiring-for-skills, firms might be limited in the extent to which they can change by the availability of skills in the labour market. Again, this is an area that has not been examined and warrants further research.

Career Planning and Retention: Nonstandard work arrangements have shifted the onus of career planning from organizations to individual workers. As individuals develop portable skills that they move from organization to organization, the traditional view of workers who build their careers in a single organization has eroded significantly (Ashford et al., 2008). Nonstandard work arrangements have also affected the organization's ability to retain standard workers. The studies examining the blended workforce suggest that a key issue in the retention of standard and nonstandard workers is how one manages the social integration of these two groups of workers (Davis-Blake et al., 2003; George, 2003). If firms are able to integrate nonstandard workers in such a way that they do not threaten the security of standard workers (George et al., 2012; Von Hippel & Kokokimminon, 2012), or that enable standard workers to engage in supervisory tasks (George, 2003), or that reduce rather than increase the workload of standard workers (Geary, 1992) then they are more likely to retain these standard workers. In addition when organizations offer employees the opportunity to shift from full-time to part-time employment, the presence of these "retention part-time workers" has a positive spill-over effect on their standard co-workers (Broschak & Davis-Blake, 2006).

SELF CHECK 7.1-1

1. A basic communication standard wherein if communication isn't complete and accurate, it can cause confusion instead of clarity.
    a. Engage in difficult conversations when necessary
    b. Let others talk
    c. Provide clear information
2. A basic communication wherein to effectively share information with another person, you have to hear what is being communicated.
    a. Let others talk
    b. Listen
    c. Engage in difficult conversations when necessary
3. A basic communication wherein it is a good way to verify what you hear so you respond appropriately.
    a. Ask questions
    b. Listen
    c. Engage in difficult conversations when necessary
4. A basic communication wherein a conversation is a two way event at a minimum.
    a. Let others talk
    b. Listen
    c. Engage in difficult conversations when necessary
5. It is said that our facial expression/s can affect the "meaning" of our words. Facial expression contributes only 1% of the message.
    a. True
    b. False
    c. None of the above

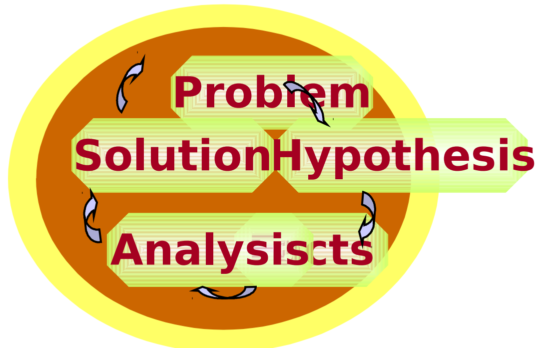
ANSWER KEY 7.1-1

1. C
2. B
3. A
4. A
5. B

INFORMATION SHEET 7.1-2

Critical Thinking

Analytical Thinking follows the scientific approach to problem solving



Defining the Problem

Definition:

- A problem is a situation that is judged as something that needs to be corrected – implies that a state of "wholeness" does not exist

Importance:

- It is our job to make sure we're solving the right problem – it may not be the one presented to us by the client. What do we really need to solve?

Basic Concepts:

- Most of the problems are initially identified by our clients
- Defining the problem clearly improves focus – it drives the analytical process
- Getting to a clearly defined problem is often discovery driven – Start with a conceptual definition and through analysis (root cause, impact analysis, etc.) you shape and redefine the problem in terms of issues

Formulating the Hypotheses

Definition:

- Hypothesis is a tentative explanation for an observation that can be tested (i.e. proved or disproved) by further investigation

Importance:

- Start at the end - Figuring out the solution to the problem, i.e. "hypothesizing", before you start will help build a roadmap for approaching the problem

Basic Concepts:

- Hypotheses can be expressed as possible root causes of the problem
- Breaking down the problem into key drivers (root causes) can help formulate hypotheses

Collecting the Facts

Definition:

- Meaningful information (has merit – not false) that is qualitative (expert opinions) or quantitative (measurable performance) to your decisions

Importance:

- Gathering relevant data and information is a critical step in supporting the analyses required for proving or disproving the hypotheses

Basic Concepts:

- Know where to dig
- Know how to filter through information
- Know how to verify – Has happened in the past
- Know how to apply – Relates to what you are trying to solve

Conducting the Analysis

Definition:

- The deliberate process of breaking a problem down through the application of knowledge and various analytical techniques

Importance:

- Analysis of the facts is required to prove or disprove the hypotheses
- Analysis provides an understanding of issues and drivers behind the problem

Basic Concepts:

- It is generally better to spend more time analyzing the data and information as opposed to collecting them. The goal is to find the "golden nuggets" that quickly confirm or deny a hypothesis
- Root cause analysis, storyboarding, and force field analysis are some of many analytical techniques that can applied

Developing the Solution

Definition:

- Solutions are the final recommendations presented to our clients based on the outcomes of the hypothesis testing

Importance:

- Solutions are what our clients pay us for…

Basic Concepts:

- It is important to ensure the solution fits the client – solutions are useless if they cannot be implemented
- Running an actual example through the solution is an effective way of testing the effectiveness and viability of the solution

Self Check 7.1-2

Enumerate the five scientific approaches in solving the problem and give the definition of each approach.

1.
2.
3.
4.
5.

Answer Key 7.1-2

Defining the Problem
Definition:
- A problem is a situation that is judged as something that needs to be corrected – implies that a state of "wholeness" does not exist
-

Formulating the Hypotheses
Definition:
- Hypothesis is a tentative explanation for an observation that can be tested (i.e. proved or disproved) by further investigation
-

Collecting the Facts
Definition:
- Meaningful information (has merit – not false) that is qualitative (expert opinions) or quantitative (measurable performance) to your decisions
-

Conducting the Analysis
Definition:
- The deliberate process of breaking a problem down through the application of knowledge and various analytical techniques
-

Developing the Solution
Definition:
- Solutions are the final recommendations presented to our clients based on the outcomes of the hypothesis testing

LO2. DETERMINE FUNDAMENTAL CAUSES OF THE PROBLEM

ASSESSMENT CRITERIA:

1. Possible causes are identified based on experience and the use of problem solving tools/analytical techniques.
2. Possible cause statements are developed based on findings.
3. Fundamental causes are identified per results of investigation conducted.

CONTENTS:

- Root cause of the problem
- Problem solving tools
- Cause and effect

CONDITION:

The students/learners must be provided with the following:

- Hand-outs

METHODOLOGIES:

- Lecture
- Group discussion

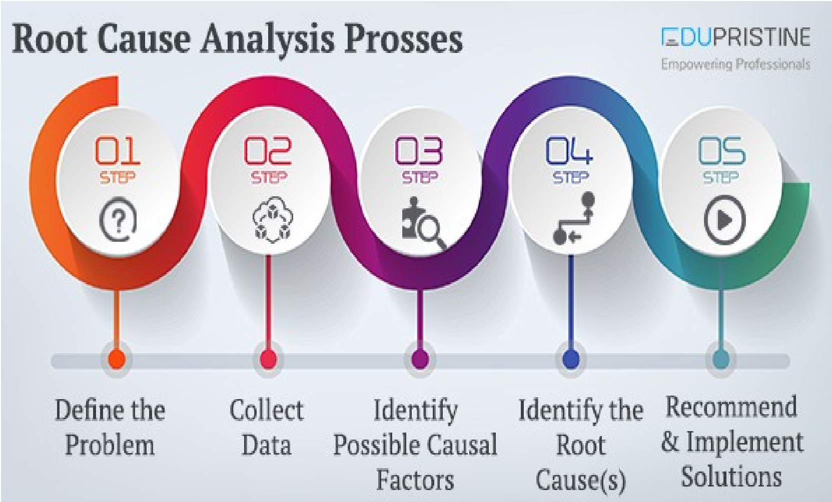
ASSESSMENT METHODS:

- Written Examination
- Oral Evaluation

INFORMATION SHEET 7.2-1

ROOT CAUSE ANALYSIS (RCA)?

A root cause is defined as a factor that caused a nonconformance and should be permanently eliminated through process improvement. The root cause is the core issue—the highest-level cause—that sets in motion the entire cause-and-effect reaction that ultimately leads to the problem(s).



Root cause analysis (RCA) is defined as a collective term that describes a wide range of approaches, tools, and techniques used to uncover causes of problems. Some RCA approaches are geared more toward identifying true root causes than others, some are more general problem-solving techniques, and others simply offer support for the core activity of root cause analysis.

- History of root cause analysis
- Approaches to root cause analysis
- Conducting root cause analysis
- Root cause analysis resources

HISTORY OF ROOT CAUSE ANALYSIS

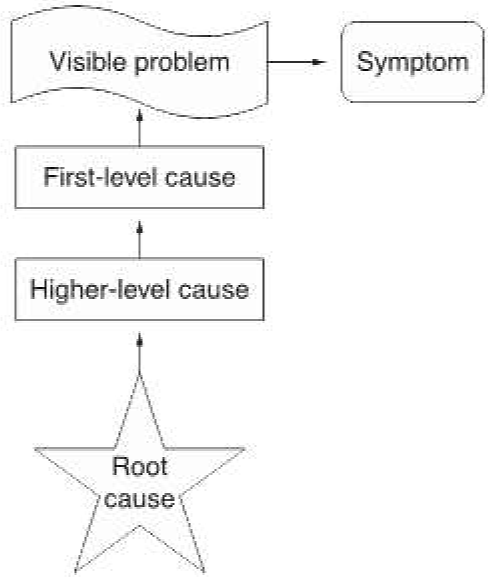
Root cause analysis can be traced to the broader field of total quality management (TQM). TQM has developed in different directions, including a number of problem analysis, problem solving, and root cause analysis.

Root cause analysis is part of a more general problem-solving process and an integral part of continuous improvement. Because of this, root cause analysis is one of the core building blocks in an organization’s continuous improvement efforts. It's important to note that root cause analysis in itself will not produce any results; it must be made part of a larger problem-solving effort for quality improvement.

APPROACHES TO ROOT CAUSE ANALYSIS

There are many methodologies, approaches, and techniques for conducting root cause analysis, including:

1. **Events and causal factor analysis:** Widely used for major, single-event problems, such as a refinery explosion, this process uses evidence gathered quickly and methodically to establish a timeline for the activities leading up to the accident. Once the timeline has been established, the causal and contributing factors can be identified.
2. **Change analysis:** This approach is applicable to situations where a system’s performance has shifted significantly. It explores changes made in people, equipment, information, and more that may have contributed to the change in performance.
3. **Barrier analysis:** This technique focuses on what controls are in place in the process to either prevent or detect a problem, and which might have failed.
4. **Management oversight and risk tree analysis:** One aspect of this approach is the use of a tree diagram to look at what occurred and why it might have occurred.
5. **Kepner-Tregoe Problem Solving and Decision Making:** This model provides four distinct phases for resolving problems:
   1. Situation analysis
   2. Problem analysis
   3. Solution analysis
   4. Potential problem analysis



Root Cause Analysis Diagram

CONDUCTING ROOT CAUSE ANALYSIS

When carrying out root cause analysis methods and processes, it's important to note:

- While many root cause analysis tools can be used by a single person, the outcome generally is better when a group of people work together to find the problem causes.
- Those ultimately responsible for removing the identified root cause(s) should be prominent members of the analysis team that sets out to uncover them.

A typical design of a root cause analysis in an organization might follow these steps:

1. A decision is made to form a small team to conduct the root cause analysis.
2. Team members are selected from the business process/area of the organization that experiences the problem. The team might be supplemented by:
   - A line manager with decision authority to implement solutions

- An internal customer from the process with problems
- A quality improvement expert in the case where the other team members have little experience with this kind of work

3. The analysis lasts about two months. During the analysis, equal emphasis is placed on defining and understanding the problem, brainstorming its possible causes, analyzing causes and effects, and devising a solution to the problem.
4. During the analysis period, the team meets at least weekly, sometimes two or three times a week. The meetings are always kept short, at maximum two hours, and since they are meant to be creative in nature, the agenda is quite loose.
5. One person in the team is assigned the role of making sure the analysis progresses, or tasks are assigned to various members of the team.
6. Once the solution has been designed and the decision to implement has been taken, it can take anywhere from a day to several months before the change is complete, depending on what is involved in the implementation process.

SELF CHECK 7.2-1

Provide the methodologies, approaches, and techniques for conducting root cause analysis:

1. ________________ Widely used for major, single-event problems, such as a refinery explosion, this process uses evidence gathered quickly and methodically to establish a timeline for the activities leading up to the accident. Once the timeline has been established, the causal and contributing factors can be identified.
2. ________________ This approach is applicable to situations where a system's performance has shifted significantly. It explores changes made in people, equipment, information, and more that may have contributed to the change in performance.
3. ________________ This technique focuses on what controls are in place in the process to either prevent or detect a problem, and which might have failed.
4. ________________ One aspect of this approach is the use of a tree diagram to look at what occurred and why it might have occurred.
5. ________________ This model provides four distinct phases for resolving problems:

ANSWER KEY 7.2-1

1. *Events and causal factor analysis*
2. *Change analysis*
3. *Barrier analysis*
4. *Management oversight and risk tree analysis*
5. *Kepner-Tregoe Problem Solving and Decision Making*

Information Sheet 7.2-2

TOOLS AND TECHNIQUES FOR CONDUCTING THE ANALYSIS

Conducting the Analysis

- The next step in problem solving is to "make sense" of the information collected in the previous step
- There is an abundance of analytical techniques that can be applied for understanding:

What are the most important issues?	Pareto Analysis
What performance areas are weak?	Benchmarking
What are the core competencies of the client?	SWOT
What forces can influence the problem?	Force Field Analysis

Analytical Techniques

- Benchmarking – Compare and measure a process or activity against an internal or external source
- SWOT Analysis – Assessment of strengths, weaknesses, opportunities, and threats
- Force Field Analysis – Overall environmental landscape and how it impacts the subject
- Cost Benefit Analysis – Compare total equivalent costs (all the minuses) against equivalent value in benefits (all the pluses)
- Impact Analysis – What if type analysis to assess the impact of change on an agency
- Pareto Chart – Bar Chart for categorizing issues or other attributes in terms of importance

Specific Sequential Steps that lead up to the Analysis

1. *Make sure you know what you are trying to solve - Clearly defined issues or questions drive the analysis!*
2. *Match up the clearly defined question or issue with the appropriate analytical tool(s)*
3. *Once you've matched up the analytical tools against the question or issue, then go out and collect the facts*

Benchmarking

- Measures and compares your performance against other similar activities or processes internally or externally
- Differences indicate possible performance issues
- May be difficult to collect comparable measurement data
- Comparing "best in class" performance is better than comparing average performance
- Best sources of data are in the private sector - Hays Benchmarking, Benchmarking Exchange, The Benchmarking Exchange, etc.

SWOT – Strengths Weaknesses Opportunities Threats

- Identifies Strengths, Weaknesses, Opportunities, and Threats by asking: What things are we good at, what things are we not good at, what things might we do, and what things should we not do?

Force Field Example

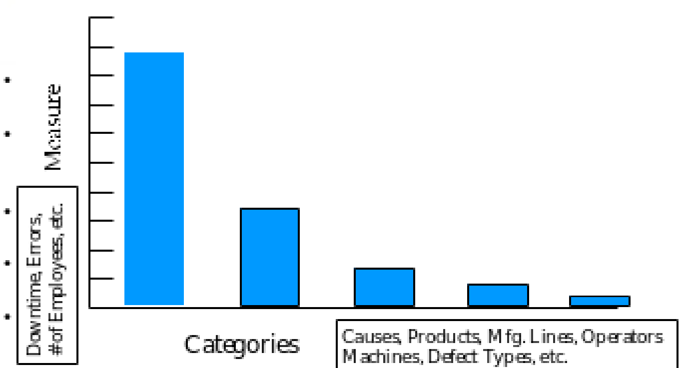
Problem: Agency is not strategically focused

Positive Forces – Promotes the Solution	Negative Forces – Inhibits the Solution

Impact Analysis

- Identifies broad and diverse effects or outcomes associated with a problem and/or the proposed solution
- Answers certain questions: How will this change impact our agency? What are the consequences of not acting on the problem?
- Objective is to minimize adverse or negative impacts going forward
- Very useful in assessing risk of different proposed solutions – helps you reach the right solution
- Numerous tools can be used to assess impacts

Pareto Chart



Impact Analysis Tools

- Scenario Playing – Storyboarding out how the future will unfold between alternatives: Do Nothing vs. Solution
- Cost Benefit Analysis - Used to quantify impacts
- Decision Tree Analysis – Build a tree and assign probabilities to each alternative to arrive at the most likely solution
- Simulation – Modeling a process and seeing how it changes when one or more variables change
- Prototype Model – Build and test the solution on a small scale before implementation to flush out lessons learned

Key Messages

- Don't rush out and collect information until you know what analytical tools you need to use – each tool has its own information needs
- Use a combination of tools to cover all the bases
- All decisions involve some assumptions – so you will never have all the facts
- Analysis is a discover driven process, it moves incrementally in baby steps – you learn, adjust and go through numerous iterations until you have insights; i.e. you can now take action on the issue or problem

Pareto Analysis

- Quantifies what is most important on a graph – 80 / 20 Rule
- Puts focus on the significant problems or issues
- Must group problems or issues based on a common and measurable attribute (such as reworks, errors, downtime, hours, etc.) = Left Vertical Axis of Bar Chart
- Must categorize problems or issues – what type is it? (poor quality, long wait times, etc.) = Right Horizontal Axis of Bar Chart
- Plot the data and rank according to frequency – descending order from left to right

Self Check 7.2-2

Give the correct analytical technique in the blank provided before the statement.

1. ________________ – Compare total equivalent costs (all the minuses) against equivalent value in benefits (all the pluses)
2. ________________ – Compare and measure a process or activity against an internal or external source
3. ________________ – Assessment of strengths, weaknesses, opportunities, and threats
4. ________________ – Bar Chart for categorizing issues or other attributes in terms of importance
5. ________________ – Overall environmental landscape and how it impacts the subject
6. _________________ – What if type analysis to assess the impact of change on an agency

Answer Key 7.2-2

1. Cost Benefit Analysis – Compare total equivalent costs (all the minuses) against equivalent value in benefits (all the pluses)
2. Benchmarking – Compare and measure a process or activity against an internal or external source
3. SWOT Analysis – Assessment of strengths, weaknesses, opportunities, and threats
4. Pareto Chart – Bar Chart for categorizing issues or other attributes in terms of importance
5. Force Field Analysis – Overall environmental landscape and how it impacts the subject
6. Impact Analysis – What if type analysis to assess the impact of change on an agency

LO3. DETERMINE CORRECTIVE ACTION

ASSESSMENT CRITERIA:

1. All possible options are considered for resolution of the problem
2. Strengths and weaknesses of possible options are considered
3. Corrective actions are determined to resolve the problem and possible future causes.
4. Action plans are developed identifying measurable objectives, resource needs and timelines in accordance with safety and operating procedures.

CONTENTS:

- Identification and analysis of possible options for problem resolution
- Corrective actions
- Principles of decision making strategies and techniques
- Action plan layouts

CONDITION:

The students/learners must be provided with the following:

- Hand-outs

METHODOLOGIES:

- Lecture
- Group discussion

ASSESSMENT METHODS:

- Written Examination
- Oral Evaluation
- Observation

Information Sheet 7.3-1

TOOLS AND TECHNIQUES FOR DEVELOPING THE SOLUTION

Basic Concepts

- Select and plan the solution that has the greatest impact on solving the problem
- Use a solutions rating matrix to weigh different solutions based on selection criteria (costs, probability of success, ease of implementation)
- Solutions should have support from your previous analysis that you can clearly communicate to the client
- Test your solutions as much as you can – use some of the Impact Analysis Tools

Common Land Mines that Blow Analytical Thinking Apart

- Once a problem is defined, Professional Consultants must have some ability to develop a possible solution. If the Consultant has **no control** to make recommendations for a problem, then the problem has been defined **outside the scope** of the project.
- The **client's definition** of the problem may **not** be **correct**. The client may lack the knowledge and experience that Professional Consultants have.
- Since most **problems** are **not unique**, Professional Consultants may be able to validate the problem and possible solutions against other sources (past projects, other experts, etc.).
- The **best solutions** to a problem are often **too difficult** for the client **to implement**. So be careful about recommending the optimal solution to a problem. Most solutions require some degree of compromise for implementation.

Key Messages

- 100% out-of-the box solutions don't exist
- No solution is a guarantee – be flexible with implementation and be willing to revisit your requirements
- Solutions rarely work unless you get buy-in and commitment from the client – if the client refuses to accept the solution, it will not work!
- Be prepared to back up your solution with an implementation plan, complete with milestones to measure performance

Summary

- Analytical Thinking follows the Scientific Approach
- Five Step Process for Consultants:
  - Define the Problem
  - Test in the form of Hypothesis
  - Focus on Facts
  - Analysis (Various Analytical Tools)
  - Recommend a Solution

Self Check 7.3-1

Give the five step process for consultants:

1.
2.
3.
4.
5.

Answer Key 7.3-1

1. Define the Problem
2. Test in the form of Hypothesis
3. Focus on Facts
4. Analysis (Various Analytical Tools)
5. Recommend a Solution

LO4. PROVIDE RECOMMENDATION/S TO MANAGER

ASSESSMENT CRITERIA:

1. Report on recommendations are prepared
2. Recommendations are presented to appropriate personnel
3. Recommendations are followed-up, if required.

CONTENTS:

- Range of formal problem solving techniques
- Sample recommendation report

CONDITION:

The students/learners must be provided with the following:

- Hand-outs

METHODOLOGIES:

- Lecture
- Group discussion

ASSESSMENT METHODS:

- Written Examination
- Oral Evaluation
- Presentation

Activity Sheet 7.4-1

Recognizing and Communicating an Emergency

Objective: The trainees will seek appropriate help in an emergency situation.

Criteria for Success: The trainees will demonstrate how to communicate an emergency.

trainees will locate and write down other important telephone numbers such as the Hospital, Police, Rescue Squad, and Poison Control.

Materials: Local telephone books. Ask students to bring their telephone books to class. They will be able to mark pages as they wish, and each student will have a book to use. Another possible source may be your workplace if you work somewhere that recycles large numbers of telephone books every year.

Paper and pens or pencils, and colored pencils.

Toy telephones or telephones that can be unplugged and therefore dialed.

Resources: Local fire, police, and emergency medical technicians may be asked to speak to the class. Field trips to any of these agencies may also be interesting and helpful to class members.

Methods: Matching, Demonstration, Drawing, Discussion, Role Play, Journal Work

Introduction: a Drama. Demonstrate crisis events, and act to the hilt. Take a doll to class and pretend that the "child" is not breathing. Feign a "heart attack." (*Tell* the class that you are going to play-act before you do the heart attack act.) Bring a toy gun or rubber knife to class and have someone in the class "attack and rob" you. Bring two matchbook cars to class and have a "wreck" with the driver (class member) of the other car. Ask the class what we call this type of event. You may get generic or specific answers: an emergency, a crisis, a wreck, a heart attack, a robbery, or you may be able only to elicit that the event is bad. The purpose of this lesson is to equip students to handle bad events effectively and promptly.

Calling for Help. Ask class members to open their telephone books. Find the emergency numbers. Typically, emergency numbers will be given on the first page or the inside cover. Identify the word emergency, and help students memorize the words fire, police, and ambulance. Discuss the symbols that identify each. Start a vocabulary list on the board with the words *emergency, fire, police, and ambulance*.

Discussion. The trainees must give examples of situations in which they would need help. List these on the board. Eventually you will need to go through the list with the class to see if there are some situations for which it would **not** be appropriate/necessary to call 911. Some learners may do better with this activity by drawing situations and then getting help identifying the right words from classmates and you. Here's a list of situations to start you thinking.

You have a fire in your kitchen. You are unable to put it out.
There is a bad wreck in front of your house. People are hurt and bleeding a lot.
There is a fire in your apartment building. You know that a man who uses a wheelchair is in his apartment on the floor above the fire.
Your co-worker falls on the stairs. He can't get up and his leg is probably broken.
Your friend trips and falls on the deck. She can walk but says her ankle hurts. It is swelling.
Your wife is going to have a baby. She thinks that it is time to go to the hospital.
Your child swallows ant poison that you have under the sink.
You cut your hand while working on the lawnmower. The cut is deep and bleeding a lot.
You burn your wrist on the car radiator. The burn is small.
Your child tries to pet the dog next door. The dog bites her and breaks the skin.
Your boss complains of numbness in his arm and pain in his chest. He is having trouble breathing.
You slip on a wet floor and fall. You are bruised and scratched.
A car runs into a telephone pole in your front yard. The driver does not answer you and does not move.

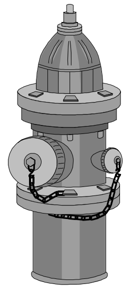
You come home after a trip to the store and find that your house has been robbed.

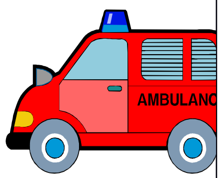
Role Play. Pick a situation with the class. Have a student volunteer to "call" 911 and ask for help. You play the 911 operator and ask the questions. For example, What has happened? What is your address? Phone number? Please stay on the phone. I am sending you (name the emergency service or just say help). You may want to ask the caller to hang up and go outside to the curb to wait for, and wave down the fire truck, EMS, etc.

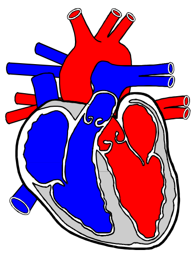
Practice with several different situations and students. Teach your students to give their name, address, phone number, and perhaps to say "I don't speak English. I speak ____." Also make sure that students understand that they must give the location and telephone number where help needs to come. They should stay on the line unless they ask the operator if they may hang up. If calling from a wireline (as opposed to a cellular) phone, 911 will have a location and send help, so that type of telephone may be preferable in an emergency situation when choices are available.

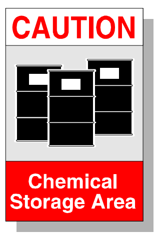
Which situations that you and the class members have listed can best be handled without a 911 call? What should you do or where should you call in these situations? See the blue (probably) pages in your telephone book for city government offices. Write the telephone number and name of the agency that you would use for each situation that is <u>not</u> a 911 call.

Activity Sheet 7.4-1

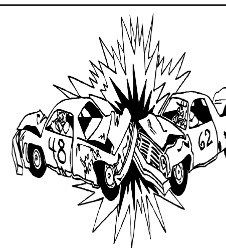


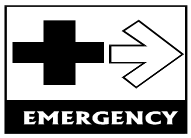


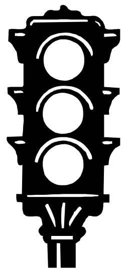


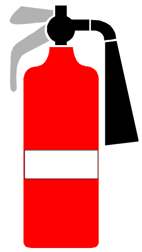


Free
Space




EMERGENCY





Place the picture beside the word or words that apply to it.

Wear your seatbelt		Policeman	
Police car		Heart Attack	
Fireman		Poison	
Emergency Medical Services		Beware the dog	
Catches fire easily		Pharmacy	
Wreck		**Stop light**	
Go out		Stop sign	
Walk carefully		Danger from electricity	
Fire hydrant		Ambulance	

Dangerous Chemicals		Police	
Go out here		Handicapped parking only	
Fire extinguisher		Emergency Room	